

Molecular Modeling of Organic and Biomolecular Systems Using *BOSS* and *MCPRO*

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Received 31 January 2005; Accepted 21 March 2005

DOI 10.1002/jcc.20297

Published online in Wiley InterScience (www.interscience.wiley.com).

Abstract: An overview is provided of the capabilities for the current versions of the *BOSS* and *MCPRO* programs for molecular modeling of organic and biomolecular systems. Recent applications are noted, particularly for QM/MM studies of organic and enzymatic reactions and for protein–ligand binding.

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Key words: BOSS; MCPRO; molecular modeling; conformational search; QM/MM; Monte Carlo statistical mechanics; enzymatic reactions; protein–ligand binding

Overview

BOSS

BOSS (biochemical and organic simulation system) is a general purpose molecular modeling system that performs molecular mechanics calculations, semiempirical molecular orbital (MO) calculations, Monte Carlo (MC) statistical mechanics simulations, and mixed quantum and molecular mechanics (QM/MM) calculations. The current distributed version is *BOSS* 4.6; the program's roots go back to MC software written by W. Jorgensen for simulation of pure liquids in the late 1970s. The molecular mechanics is usually carried out for isolated molecules or complexes in the gas phase, and covers energy minimizations, normal mode analysis, and conformational searching. The MC simulations can be executed for pure liquids, solutions, clusters, or gas-phase systems; typical applications include computing the structure and thermodynamic properties of a liquid at a specified temperature and pressure, free energies of solvation, effects of solvation on relative energies of conformers, locations of transition structures for reactions in solution, changes in free energies of solvation along reaction paths, and structures and relative free energies of binding for host–guest complexes in solution. The energy of the system is represented by the OPLS (optimized potentials for liquid simulations) force fields^{1,2} or through QM/MM calculations³ using the semiempirical QM methods, AM1,⁴ PM3,^{5,6} or PDDG/PM3.^{7,8} Solute molecules are normally represented as Z-matrices, which designate the atomic numbers and bond lengths, bond angles, and dihedral angles (Fig. 1). Molecules can be fully or partially flexible by specifying which internal coordinates are variable in the input Z-matrix. Coordinate files are output in Protein Data Bank (PDB)

format for easy display with molecular graphics programs. As an alternative to inclusion of explicit solvent molecules, a version⁹ of the GB/SA continuum solvation model,¹⁰ which has been optimized for the OPLS all-atom force field (OPLS-AA), has also been implemented for use in the MM and MC calculations.

Some important features of the molecular mechanics calculations are: (1) multiple optimizers are provided for energy minimizations including nongradient methods (Simplex, Powell, and Hooke–Jeeves), gradient methods (Fletcher–Powell, BFGS, and conjugate-gradient), and simulated annealing. The optimal choice of method depends on the nature of the energy surface. (2) Repetitive energy minimizations can be performed automatically as one internal degree of freedom (bond length, bond angle, or dihedral angle) is used as a reaction coordinate. For a dihedral angle, this is referred to as “dihedral angle driving.” (3) Vibrational frequencies and statistical thermodynamic properties such as zero-point energy and gas-phase heat capacity and entropy can be computed from a normal coordinate analysis. (4) Conformational searches for multiple energy minima are performed automatically by an internal coordinate Monte Carlo procedure¹¹ or by stochastic search.¹² Exploration of low-energy regions of conformational space in the MC approach is enhanced by choosing starting geometries from among previously found low-energy conformers.

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Contract/grant sponsor: National Science Foundation; contract/grant number: CHE-0130996

Contract/grant sponsor: National Institutes of Health; contract/grant number: GM-32136

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ethanol/ethyl chloride FEP
 1 OCL 154 151 0 0.000000 0 0.000000 0 0.000000
 2 DUM -1 0 1 0.500000 0 0.000000 0 0.000000
 3 DUM -1 0 2 0.500000 1 90.000000 0 0.000000
 4 H 155 100 1 0.946000 2 90.000000 3 180.000000
 5 Cl 157 152 1 1.410000 4 109.200000 2 180.000000
 6 C2 135 0 5 1.530000 1 108.900000 4 180.000000
 7 HC 140 153 5 1.091000 6 110.500000 1 120.500000
 8 HC 140 153 5 1.091000 6 110.500000 1 239.500000
 9 HC 140 0 6 1.091000 5 111.000000 1 180.000000
10 HC 140 0 6 1.091000 5 111.000000 9 120.000000
11 HC 140 0 6 1.091000 5 111.000000 9 240.000000
      Geometry Variations follow
00040001 0.30
00050001 1.78
      Variable Bonds follow
0004-0011
      Additional Bonds follow
      Harmonic Restraints follow
      Variable Bond Angles follow
0005-0011
      Additional Bond Angles follow
AUTO
      Variable Dihedrals follow
0006-0011
      Additional Dihedrals follow
AUTO
      Domain Definitions follow
      Conformational Search
      Final blank line

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Figure 1. An example of a *BOSS/MCPRO* Z-matrix for conversion of ethanol to ethyl chloride. Atom 1 is the oxygen that becomes chlorine and changes from OPLS-AA atom type 154 to 151. The geometry for ethanol ($\lambda = 0$) is given in the main body of the Z-matrix, and the modifications needed to covert it to ethyl chloride ($\lambda = 1$) are specified under "Geometry Variations." For example, atom 4, the hydroxyl hydrogen, becomes a null atom with type 100 and its bond shrinks to 0.3 Å. Only atoms whose type changes need specification of a nonzero final type in column 4. Atoms 2 and 3 are examples of type -1 dummy atoms that may be used to facilitate construction of a Z-matrix. Column 2 is just a label for printing; the "two-letter" codes are determined by the atom types in columns 3 and 4. If there are additional solutes, they are separated by a line containing "TERZ" and the atom numbering in column 1 continues.

Some important features of the Monte Carlo simulations are: (1) simulations are carried out at any specified temperature and at either constant pressure (NPT) or constant volume (NVT) with the Metropolis algorithm.¹³ Calculations are performed with either periodic boundary conditions or for a cluster of molecules and/or ions. (2) Both solute and solvent molecules can be rigid or flexible. Variations in internal coordinates are performed for the flexible degrees of freedom. Sampling also includes random translations and rigid-body rotations. Water molecules are normally treated as rigid bodies that only translate and rotate. (3) For periodic systems, long-range electrostatic interactions can be evaluated by the Ewald method.¹³ Otherwise, spherical truncation of the interaction energies is performed with quadratic smoothing at the cutoffs. The cutoffs are typically based on heavy atom–heavy atom separations in a "residue-based" manner; that is, the interacting molecular fragments are normally neutral or they can have integral charges for ions. For example, the decision to include a water–water interaction is based on the O–O distance, and if it is below the cutoff, the interaction between all atoms is evaluated. (4) Extensive capabilities are provided for easy setup of the solvent molecules from equilibrated boxes of 10 common organic solvents plus TIP3P, TIP4P, and TIP5P water^{14,15} or from user-provided solvent

boxes that are created by running a pure-liquid simulation with *BOSS*. Binary solvent mixtures are also readily treated. (5) Free energy changes are computed from statistical perturbation (FEP) theory in a windowing format.^{16–19} With use of the internal coordinates for the solutes, it is straightforward to obtain a free energy profile as a function of a specified dihedral angle or intermolecular distance. Computation of two-dimensional free-energy maps is also routine by repetitive calculation of one-dimensional profiles.

A large library of scripts is provided for facile execution of standard procedures. For example, given a Z-matrix *zmat*, the command *xCSGB500 zmat* causes MC conformational search to be performed using the OPLS-AA force field with GB/SA hydration and 500 starting structures. *xPDGOPT zmat* is used to perform a PDDG/PM3 optimization for the system specified in *zmat*. *xFEP-DMSO zmat* causes a MC/FEP calculation to be run using periodic boundary conditions, 10 million configurations of averaging following 2 million of equilibration for each of 10 windows, with 260 DMSO (dimethylsulfoxide) solvent molecules at 25°C and 1 atm. Other scripts perform utility operations such as converting an input PDB (*xPDBZ*) or mol-format (*xMOLZ*) file to a *BOSS/MCPRO* Z-matrix.

MCPRO

MCPRO (Monte Carlo for proteins) stemmed from *BOSS* and retains most of its features and capabilities. However, *MCPRO* is designed for efficient MC simulations of biomolecules in solution, and it makes extensive use of the concept of residues. Thus, residue–residue and solvent–residue lists are maintained to facilitate the energy calculations. A utility program *pepz* allows easy system setup starting from a PDB file or *de novo* creation of any polypeptide or polynucleotide. For energy minimizations, several procedures are available, although conjugate gradient with analytical derivatives is usually the most efficient one. For the MC simulations, the NPT or NVT ensembles can again be selected and the solvent can be explicit or approximated by GB/SA. Ewald treatment is available for long-range electrostatic interactions, although solvent spheres (caps) are often used rather than periodic boundary conditions, for example, for simulations of protein–ligand complexes.

An important recent development was methodology for protein and polynucleotide backbone sampling using a MC procedure featuring concerted rotations with variable bond angles and dihedral angles (CRA).^{20,21} Formerly, for biomolecules, the MC moves usually involved variation of side-chain bond angles and dihedral angles for one randomly picked residue at a time, while the backbone was normally held fixed after a conjugate gradient optimization. However, CRA has been implemented in *MCPRO* for the backbone sampling, and it has allowed, for example, *ab initio* polypeptide folding from extended structures using GB/SA solvation (Fig. 2).²² The principal items from *BOSS* that have not been implemented in *MCPRO* are the conformational search and normal mode calculations; *BOSS* can be used in these cases for systems with up to ca. 1000 atoms. Also, the QM/MM capabilities have been added to an in-house version of *MCPRO*, which permits study of enzymatic reactions.^{23,24} However, QM/MM is not in the latest, distributed version, *MCPRO* 2.0. The most com-

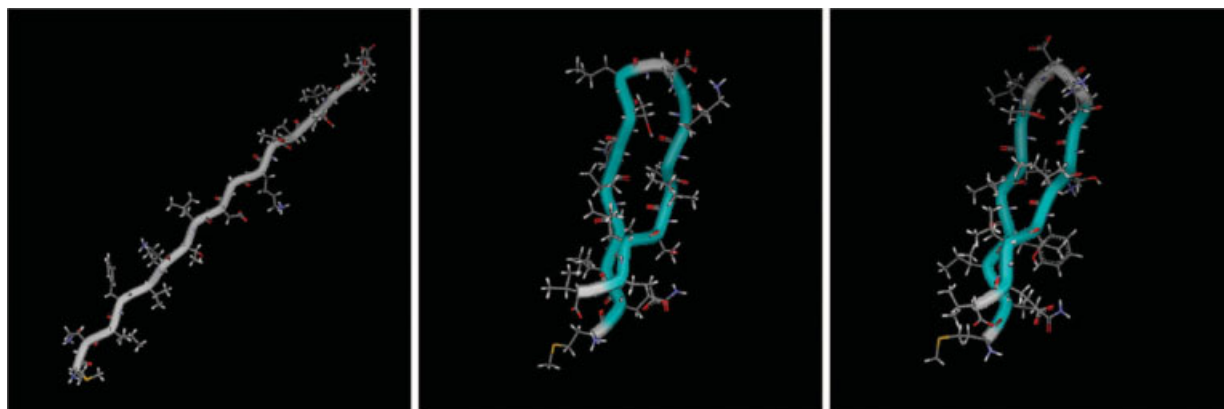


Figure 2. Folding of the U(1-17)T9D heptadecapeptide using MCPRO, the OPLS-AA force field, GB/SA hydration, and the CRA backbone sampling algorithm. Left: start of the MC run. Middle: after 24 million MC steps. Right: NMR structure 1e0q. The backbone RMSD is 2.5 Å between the computed and NMR structures. See ref. 22 for details.

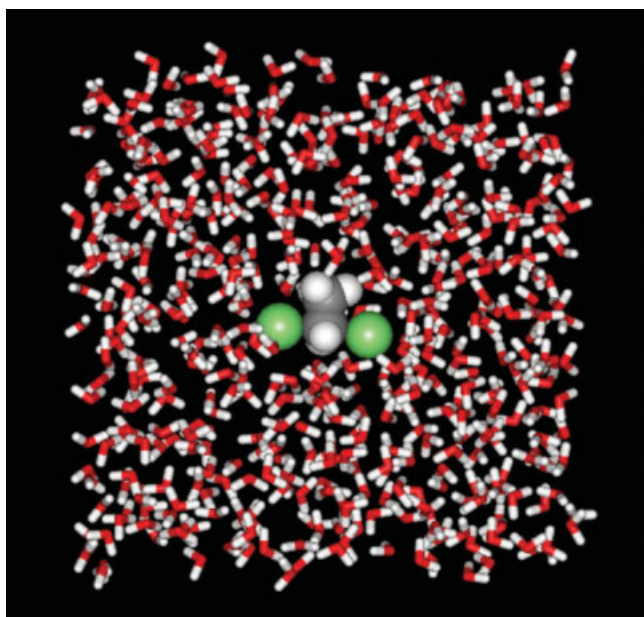


Figure 3. One configuration of the transition structure from a QM/MM MC/FEP simulation for the reaction of ethyl chloride with chloride ion in TIP4P water. See ref. 43 for more details. A few of the 500 water molecules in front of the solute have been removed for clarity.

mon application of *MCPRO* to date has been MC/FEP computation of relative free energies of binding for protein–ligand complexes; this may focus on the effects of either modifying the ligand²⁵ or the protein.²⁶

Energy and Free Energy Evaluation

The molecular mechanics calculations typically use the OPLS-AA force field,² which is nearly identical in form to other popular force

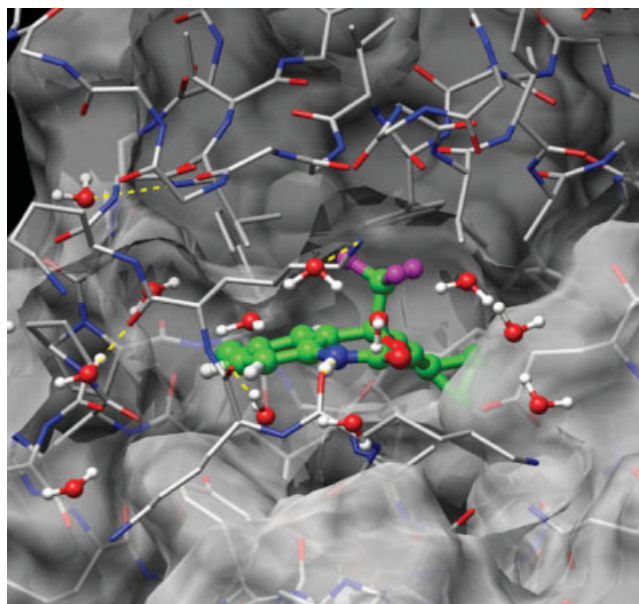


Figure 4. A configuration from a Monte Carlo simulation of efavirenz (Sustiva) bound to HIV-1 reverse transcriptase (HIV-RT). Results from MC/FEP calculations using MCPRO provided confidence in the predicted structure for the efavirenz/HIV-RT complex in ref. 50 and distinguished between two alternative X-ray structures for the complex of efavirenz with the clinically important K103N mutant of HIV-RT in ref. 25.

fields such as AMBER and CHARMM.^{27,28} The energy expression consists of harmonic terms for bond stretching and angle bending, a Fourier series for each torsional angle, and Coulomb and Lennard–Jones interactions between atoms separated by three or more bonds [eqs. (1)–(4)]. The latter “nonbonded” interactions are also evaluated between intermolecular atom pairs to yield the intermolecular energy, and they are reduced by a factor of 2 for intramolecular 1,4-interactions. Geometric combining rules are

used for the Lennard–Jones parameters, that is, $\sigma_{ij} = (\sigma_i\sigma_j)^{1/2}$ and $\varepsilon_{ij} = (\varepsilon_i\varepsilon_j)^{1/2}$. The total energy then consists of these terms evaluated for all particles in the system; optional, additional terms are (1) harmonic restraints on interatomic distances, dihedral angles, or for an atom to a point in space, (2) flat-bottomed harmonic

$$E_{\text{bond}} = \sum_i k_{b,i}(r_i - r_{0,i})^2, \quad (1)$$

$$E_{\text{bend}} = \sum_i k_{\theta,i}(\theta_i - \theta_{0,i})^2, \quad (2)$$

$$E_{\text{torsion}} = \sum_i \{V_{1,i}(1 + \cos \varphi_i)/2 + V_{2,i}(1 - \cos 2\varphi_i)/2 + V_{3,i}(1 + \cos 3\varphi_i)/2 + V_{4,i}(1 - \cos 4\varphi_i)/2\}, \quad (3)$$

$$E_{nb} = \sum_{i < j} \{q_i q_j e^2 / r_{ij} + 4\varepsilon_{ij}[(\sigma_{ij}/r_{ij})^{12} - (\sigma_{ij}/r_{ij})^6]\}, \quad (4)$$

$$E_{ab} = \sum_i \sum_{j \in a}^{\in a} \sum_{j \in b}^{\in b} \{q_i q_j e^2 / r_{ij} + 4\varepsilon_{ij}[(\sigma_{ij}/r_{ij})^{12} - (\sigma_{ij}/r_{ij})^6]\}, \quad (5)$$

potentials as NOE (nuclear Overhauser effect) restraints, (3) a half-harmonic restoring potential when a solvent cap is used to inhibit solvent evaporation, (4) the Ewald energy for long-range electrostatics in periodic systems, (5) an energy correction for the Lennard–Jones interactions neglected beyond the cutoffs for non-aqueous solvents or pure liquids, and (6) the GB/SA solvation energy. There are also many other options, as designated in the “parameter file” for use in specialized cases. For example, one can perform “multiple-copy” calculations by declaring that solute–solute or solvent–solvent interactions are not to be evaluated. Thus, one could have a protein as a solute and solvate with thousands of probe molecules; then an MC simulation or energy minimization could be performed to determine where the probes cluster in the protein.²⁹

When the force field is used, the OPLS atom-type number for each atom is specified in the solute Z-matrix (Fig. 1). The corresponding parameters are extracted automatically from the *oplsaa.par* (nonbonded and torsion) and *olpsaa.sb* (bond stretch and angle bend) parameter files that are provided with the programs, and which may be edited to add new parameters. The specification of the nonbonded parameters in *oplsaa.par* covers the atomic charge q , the Lennard–Jones parameters, σ and ε , and designation of a general atom type, for example, CT for aliphatic carbon or NZ for a cyano nitrogen. These two-letter codes are then used for retrieval of the bond stretching, angle bending, and torsion parameters. Thus, a calculation for butane requires CT–CT and CT–HC stretching, CT–CT–CT, HC–CT–CT, and HC–CT–HC bending, and HC–CT–CT–HC, HC–CT–CT–CT, and CT–CT–CT–CT torsion parameters. If any parameters are missing, warnings are issued, and synonym and estimation procedures, based on the two-letter codes, are invoked to provide estimated values. Improper torsions, which are often used to hinder pyramidalization at

trigonal centers, are implemented by designating nonzero V_2 terms in eq. (3).

The Monte Carlo sampling in *BOSS* and *MCPRO* uses standard procedures including the Metropolis algorithm.¹³ For a periodic system of flexible molecules, an attempted MC move consists of randomly picking one molecule, translating it randomly in all three Cartesian directions, randomly rotating it about one randomly chosen Cartesian axis, and rebuilding the molecule from its Z-matrix with random changes to any designated bond lengths, bond angles, and dihedral angles. The ranges for the random variations are adjusted during the equilibration phase to give acceptance probabilities for new configurations of ca. 40%. This procedure does not introduce nonunitary Jacobian terms in the MC transition probabilities that could be introduced, for example, by use of spherical coordinates or Euler angles. This is verified by performing MC simulations of purely harmonic systems such as a triatomic with two bond stretching and one angle bending terms; the computed potential energy in this case must agree with the equipartition result of $(3/2)k_B T$. For MC simulations in the NPT ensemble, attempted changes to the volume of the periodic system are also made periodically. This can be done by randomly scaling the positions of all molecules outward or inward. In this case, the Metropolis test features comparison of $\exp[-(E + PV - Nk_B T \ln V)/k_B T]$ for the new and old configurations.¹³

Atomic Charges and the OPLS/CM1A Force Field

OPLS-AA parameters have been explicitly developed for many organic functional groups, and missing bond stretching, angle bending, torsion, and Lennard–Jones parameters can usually be well estimated. For an arbitrary new molecule with a functional group or heterocycle that does not have OPLS-AA parameters, the principal problem is then the assignment of the partial atomic charges, q , for eqs. (4) and (5). The most common solution has been to perform a QM calculation and to fit the charges to reproduce electronic properties, especially dipole moments and electrostatic potentials. This has led to several common alternatives such as CHLPG and RESP charges that are usually derived from *ab initio* wave functions, and faster approaches using, for example, semiempirical QM charges.³⁰ Our own preference has been for the CM charges of Cramer, Truhlar, and coworkers, which are derived from AM1 or PM3 calculations.^{31,32} Their rapid procedures, which yield highly accurate dipole moments, are incorporated directly into *BOSS* and *MCPRO* for computing charges for arbitrary organic molecules for both force-field and QM/MM calculations. Some symmetrization of the CM charges is performed such that the charges for monovalent atoms Y in XY_2 or XY_3 groups are made to be the same, as are charges for equivalent carbons and hydrogens in substituted benzenes. Without this modification, for example, QM charges for the ortho hydrogens in phenol would be different. This is particularly undesirable for conformational searching, because it leads to structurally equivalent minima with different energies.

The CM alternatives have been tested in computations of free energies of hydration in TIP4P water.³³ The best results come from the AM1-based CM1A charges scaled by 1.14 for neutral mole-

Table 1. Representative Conformational Energy Differences (kcal/mol).

Molecule	Conformers	Δ OPLS-AA	Δ E OPLS/CM1A ^a	Δ Exptl. ^b
butane	gauche-anti	0.80	0.44	0.75
methylcyclohexane	axial-equat.	1.78	1.25	1.75
1,2-dichloroethane	gauche-anti	1.11	1.08	1.20
propanal	skew-syn	1.39	1.49	0.95
ethyl methyl ether	gauche-anti	1.47	1.20	1.50
propylamine	gauche-anti	0.54	0.26	0.42
N-methylpiperidine	axial-equat.	3.60	3.24	3.15
N-methylacetamide	E-Z	2.30	1.50	2.30
glyoxylic acid	s-trans E-Z	1.25	2.05	1.20
2-methoxy-THP	equat.-axial	1.34	2.37	1.05

^aUsing charges computed for the lower-energy conformer.^bRef. 34.

cules. They yield an average error of 1.0 kcal/mol for 25 diverse organic molecules, while the error is 0.7 kcal/mol with OPLS-AA charges.³³ For ions, the CM charges are not scaled to retain integer charges. Through use of the CM1A charges, it is possible to readily perform force-field calculations for essentially any molecule that has AM1 and OPLS-AA parameters, which includes elements H, B, C, N, O, F, Si, P, S, Cl, Br, and I. The charges come from CM1A, the two-letter codes are assigned, and then the stretching, bending, and torsion parameters are taken from OPLS-AA. Scripts are again provided to facilitate the process starting from common file types; for example, *xPDBZ xxx* converts the PDB file *xxx.pdb* to a BOSS/MCPRO Z-matrix *xxx.z* with the scaled CM1A and nonbonded parameters appended. Though the resulting general force field, OPLS/CM1A, is not as accurate as OPLS-AA, the degradation is modest; some conformational results are compared in Table 1.

Semiempirical QM Methods

The choice of QM methods has been made with QM/MM calculations in mind.³⁵ Our interests have not been in just performing energy minimizations; rather, the aim has been to incorporate on-the-fly QM in MC or molecular dynamics (MD) simulations. In this manner, one can explore the free-energetics and mechanisms for reactions in solution and catalyzed by enzymes. In view of the desired system sizes, semiempirical QM (SQM) is the only realistic choice at this time.³⁶ In particular, the key aim is to quantify activation barriers and stationary points on the reaction paths; this requires free-energy calculations and extensive configurational sampling via MC or MD. In our QM/MM calculations for Diels–Alder reactions in solution, computation of each free energy profile required 3.5 million single-point QM calculations.³⁷ However, a single distance sufficed as the reaction coordinate in that case. In all of our subsequent work on pericyclic, elimination and substitution reactions, for example, recent S_NAr and Kemp decarboxylation studies,³⁸ it has been necessary to compute two-dimensional free-energy maps, each requiring ca. 50 million single-point QM calculations. For even moderate sized systems with ca. 20

heavy atoms, SQM single-point calculations are 1000 to 10,000 times faster than DFT calculations such as B3LYP/6-31G* or moderate basis-set Hartree–Fock calculations, for example, HF/6-31G*, and 500,000 times faster than MP2/6-31G*.

Thus, the well-known, MNDO-based AM1 and PM3 methods^{4–6} have been implemented in our software, and the development of improved SQM methods has also been undertaken. Following success with improving computed ΔH_f s with bond and group adjustments,³⁹ related modifications to the MNDO core repulsion formula (CRF) were pursued. The principal difference between AM1 and MNDO is the addition in AM1 of numerous Gaussians to the MNDO CRF, for example, seven Gaussians for a C–H interaction.³ Starting with MNDO, we instead added the simpler pairwise distance directed Gaussian expression in eq. (6), which uses four terms for an A–B interaction and three for A–A. This method, PDDG/MNDO, is simpler than AM1, but upon optimization reduced the mean

$$PDDG(A, B) = \frac{1}{z_A + z_B} \times \left[\sum_{i=1}^2 \sum_{j=1}^2 (P_{Ai} + P_{Bj}) \exp(-10(R_{AB} - D_{Ai} - D_{Bj})^2) \right], \quad (6)$$

unsigned error (mue) for heats of formation (ΔH_f) for 622 CHNO compounds from 8.4 (MNDO) and 6.7 (AM1) to 5.2 kcal/mol (PDDG/MNDO).⁷ The mue with PM3 for the 622 compounds is notably only 4.4 kcal/mol. This was surpassed by adding the PDDG terms to PM3 yielding PDDG/PM3 and a mue of 3.2 kcal/mol.⁷ In the process, well-known problems with homologation and hydrocarbon branching were corrected, and results for diverse isomerization energies were shown to be more accurate with PDDG/PM3 (mue = 1.6) than B3LYP/6-31G* (mue = 2.3 kcal/mol). ΔH_f results and, therefore, heats of reaction are greatly improved with PDDG/PM3 over B3LYP methods; for example, for the G3 dataset, the mues are 3.2 for PDDG/PM3 and 7.2 kcal/mol for B3LYP/6-311+G(3df,2p).⁷

The PDDG methods were then parameterized for halogens⁸ and dramatic improvements were obtained for the ΔH_f mues of 442 F, Cl, Br, and I-containing compounds: MNDO (14.0), AM1 (11.1), PM3 (8.1), PDDG/MNDO (6.6), and PDDG/PM3 (5.6 kcal/mol). The error here for PDDG/PM3 comes predominantly from polyfluoro compounds. Hydrogen-bonded and ion-molecule complexes and transition states were also in the training set; the results are similar to those from B3LYP/6-311++G(d,p). These items were not in the original CHNO training set,⁷ and although the results are better than for AM1, significant improvements were not made in these areas over PM3. The PDDG/PM3 parameterization for Si, P, and S has also been completed with particularly large improvements for sulfur compounds including hypervalent ones.⁴⁰ ΔH_f mues for 249 S-containing compounds are MNDO (41.2), AM1 (10.6), PM3 (10.5), MNDO/d (10.0), and PDDG/PM3 (6.4). Current work is focusing on reintroduction of the overlap integrals in the secular determinant; in particular, good success has been obtained with a nonorthogonal PM3 method with simultaneous parameterization for ground and transition states.

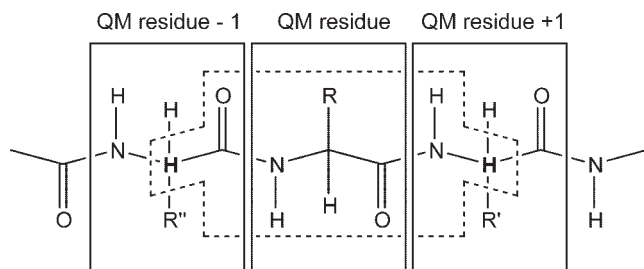
QM/MM Calculations

The implemented QM/MM methodology allows designation of multiple solute molecules and/or residues of biomolecules to be treated by the SQM methods and the remainder of the system by MM and/or GB/SA [eq. (7)].³ For the typical case of QM solutes

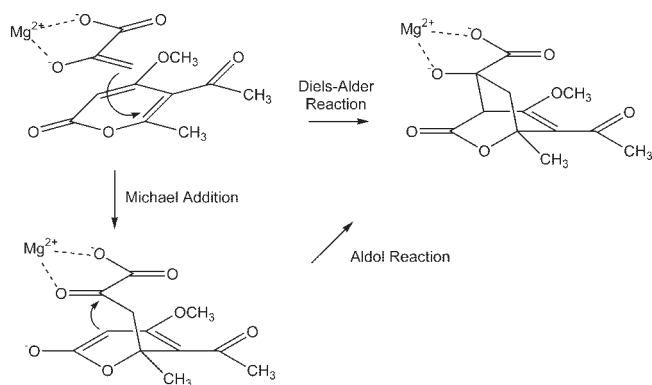
$$E_{\text{total}} = E_{\text{QM}} + E_{\text{QM/MM}} + E_{\text{MM}}, \quad (7)$$

$$E_{\text{QM/MM}} = \sum_{i \in \text{QM}} \sum_{j \in \text{MM}} \{q_i q_j e^2 / r_{ij} + 4\epsilon_{ij} [(\sigma_{ij}/r_{ij})^{12} - (\sigma_{ij}/r_{ij})^6]\}, \quad (8)$$

in a MM solvent, the QM/MM energy is given by the non-bonded terms in eq. (8), where the q_i for the QM atoms are the CM1 charges (scaled by 1.14 for neutral molecules) and the q_j are OPLS-AA charges, and standard OPLS-AA Lennard-Jones parameters are used based on the two-letter codes. In an MC simulation, on each attempted move of a solute, the QM energy and CM charges are computed for the new structure and they are used in the new E_{total} . The old density matrix is used in the first SCF cycle to speed convergence. Furthermore, for an MC/FEP calculation, if the attempted move is accepted, the QM



Scheme 1



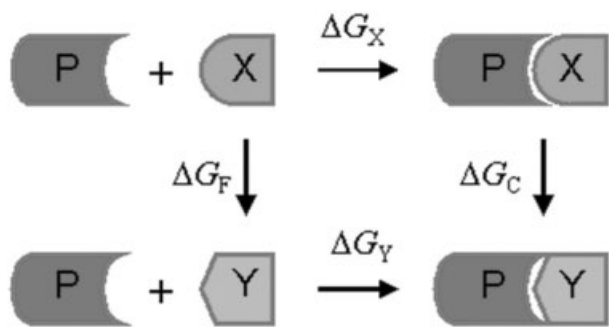
Scheme 2

and CM calculations are then performed for the perturbed versions of the solutes to evaluate E_{total} for the perturbed systems. For enzymatic reactions in which a subset of the amino acid residues is treated by QM, a link-atom procedure is used to meld the QM and MM parts of the protein.²⁴ The partition is performed along the backbone; the link atoms are hydrogens with bonds to the QM region elongated by ca. 0.5 Å so that the hydrogens and C_α atoms overlap (Scheme 1). In this manner, the link atom and its C_α form constitute a single particle that works collectively to interact with the system.

Results of QM/MM calculations using *BOSS* or *MCPRO* have been reported that well reproduce observed free energies of hydration for 25 diverse organic molecules,³³ and solvent effects for conformational equilibria,³ the parent Claisen rearrangement,³ three Diels-Alder reactions,³⁷ the chorismate to prephenate Claisen rearrangement in water, methanol, and catalyzed by chorismate mutase,^{41,42} S_N2 reactions (Fig. 3),⁴³ the $N_3^- + p\text{-FPhNO}_2$ S_NAr reaction and Kemp decarboxylations.³⁸ A recent study clarified the mechanism for the reaction of 2-pyrone derivatives catalyzed by macrophomate synthase as a tandem Michael-aldol sequence rather than a Diels-Alder reaction (Scheme 2).²⁴ Most cases involve locating the stationary points on computed two-dimensional free energy surfaces or "potentials of mean force" (pmfs). The studies for S_N2 and S_NAr reactions and the Kemp decarboxylations used PDDG/PM3 for the QM and yielded solution-phase activation barriers in excellent accord with experiment. The gas-phase results for the S_N2 reactions are virtually identical for PDDG/PM3 and CBS-QB3.⁴³ The rate retardation for S_N2 reactions on neopentyl halides was confirmed to be a steric effect in all media in contrast to a provocative experimental report.⁴³ The computations have been highly automated such that an entire reaction profile can be computed via one job submission.

Free-Energy Perturbation Calculations

The MC/FEP calculations use the Zwanzig expression [eq. (9)]¹⁸ to compute the free energy change between a reference system 0 and a perturbed system 1.¹⁶⁻¹⁹ The average is taken for sampling configurations of the reference system. For potential-of-mean-



Scheme 3

$$\Delta G(0 \rightarrow 1) = -k_B T \ln \langle \exp[-(E_1 - E_0)/k_B T] \rangle_0, \quad (9)$$

force calculations, the perturbations are for a reaction coordinate that is normally defined as an intermolecular distance or dihedral angle. Perturbations are then made progressively from a starting point r_0 using increments Δr of typically 0.02–0.05 Å for a distance or 2–5° for an angle. In practice, “double-wide sampling”¹⁹ is used such that two free-energy increments are obtained from each simulation corresponding to r_i going to $r_i + \Delta r$ and $r_i - \Delta r$, then the next simulation is run at $r_i + 2 \Delta r$. In *BOSS* and *MCPRO*, Z-matrices and Cartesian coordinates are maintained internally for the reference and the two perturbed solute systems.

For relative free energies of solvation, a perturbation is performed to convert a molecule A to molecule B both in solution and in the gas phase; the difference in the two free energy changes gives the difference in free energy of solvation.^{19,33} The corresponding reaction coordinate involves a coupling parameter, λ , and its increment, $\Delta\lambda$, that allow one molecule to be smoothly mutated to the other by changing the force field parameters and geometry.¹⁹ For binding calculations, perturbations are made to convert one ligand to another using the thermodynamic cycle in Scheme 3. The difference in free energies of binding for the ligands X and Y then comes from eq. (10). Two series of mutations are performed to convert X to Y free in the solvent and complexed to the host P, which yield ΔG_F and ΔG_C . The same cycle can be used to compute the effect of a protein mutation on the binding of one inhibitor by making P the inhibitor and X and Y the two proteins.

$$\Delta\Delta G_b = \Delta G_X - \Delta G_Y = \Delta G_F - \Delta G_C. \quad (10)$$

In the protein–ligand binding arena, MC/FEP calculations with *MCPRO* have been used in recent studies, for example, to examine complexation of cyclosporins with cyclophilins in a gene therapy context,⁴⁴ neurotrophic inhibitors with FKBP,⁴⁵ peptide mimetics with Src SH2 domain,^{46,47} substituent optimization for selective COX-2 inhibitors,⁴⁸ the origin of COX-2/COX-1 selectivity,⁴⁹ and prediction of effects of protein mutations on the activity of anti-HIV drugs.^{25,26,50,51} The latter calculations were also used to validate predictions for the structures of the complexes of efavirenz (Sustiva, Fig. 4)²⁵ and the drug candidate TMC125 with HIV-1 reverse transcriptase;²⁶

the predictions were shown to be correct by subsequent X-ray diffraction studies.^{52,53}

Linear Response

Although we emphasize FEP calculations, approximate approaches based on linear response (LR) theory have also been explored and implemented in *BOSS* and *MCPRO*. In Åqvist’s model,⁵⁴ the free energy of interaction of a solute with its environment is given by one-half the electrostatic (Coulombic) energy plus the van der Waals (Lennard–Jones) energy scaled by an empirical parameter, α . For binding a ligand to a protein, the differences in the interactions between the ligand in the unbound state and bound in the complex then provide an estimate of the free energy of binding via eq. (11). The required energy components

$$\Delta G_b = \beta \langle \Delta E_{\text{elec}} \rangle + \alpha \langle \Delta E_{\text{vdw}} \rangle, \quad (11)$$

$$\Delta G_b = \beta \langle \Delta E_{\text{elec}} \rangle + \alpha \langle \Delta E_{\text{vdw}} \rangle + \gamma \langle \Delta \text{SASA} \rangle, \quad (12)$$

were obtained from MD simulations for inhibitors in water and for the protein–inhibitor complexes in water. Key advantages over FEP methods are (a) absolute free energies of binding can be approximated, and (b) only simulations at the end points of a mutation are required. A major plus for the LR approach is that it is easy to treat structurally diverse ligands, which might be impractical to tackle with FEP calculations. In view of the approximations in eq. (11) including the neglect of intramolecular energetics, the approach is expected to be most viable for a series of relatively rigid analogs.

Early extension of the LR method to calculate free energies of hydration (ΔG_{hyd}) incorporated a third term proportional to the solute’s solvent-accessible surface area (SASA), as an index for cavity formation.⁵⁵ An analogous expression, eq. (12), was used to correlate successfully observed binding affinities for series of inhibitors with several proteins including thrombin⁵⁶ and FKBP.⁵⁷ Results have since been obtained for much larger data sets and show that better accuracy can be achieved by considering alternate descriptors, for example, for ca. 250 non-nucleoside inhibitors with HIV-RT,⁵⁸ and 148 inhibitors with CDK2, Lck, and p38 kinase.⁵⁹ In these studies, the protein–ligand interaction energy and a desolvation term involving the change in hydrogen bonding for the ligand upon binding emerge as dominant. LR methods are a merger of QSAR (quantitative structure–activity relationship) and simulation technology; they have the advantage of using powerful descriptors including energy components and estimates of solvation energies that are extracted from the MD or MC simulations on the protein–ligand complexes. Other approximate methods have also been introduced; they typically feature representation of the solvent through Poisson–Boltzmann or GB/SA procedures.

Distribution, Files, and Input

BOSS and *MCPRO* are written mostly in FORTRAN with some C, and they are normally run with command line submission of batch

Table 2. Files Used by BOSS and MCPRO.

Unit	Name	Description
IRD	CMD	The “command” file for executing the program, e.g., a C-shell or MS-DOS script.
ILST	OT	The printed output “ot” file lists parameters for the calculation, sampling information, and structural and energetic results.
IDSKI	IN	The “in file” contains the coordinates and other data needed to restart a MC simulation.
IDSKO	UP	The “up file” contains the results for each MC run that are needed to compute the global averages.
IDSKS	SV	The “save file” is used to store all coordinates periodically for later analysis of MC runs.
IDSKA	AV	The “av file” contains output distribution functions that may be plotted.
IZM	ZMAT	Contains the input Z-matrix for the solutes.
IZMS	ZMAT	Optional Z-matrix for a custom, user-input solvent.
IPAR	PAR	The “par” file contains details and options for the present calculation such as temperature, pressure, choice of optimizer, inclusion of Ewald, etc. The <i>oplsaa.par</i> file is appended and contains the OPLS-AA non-bonded and torsion parameters.
IBARPAR	<i>oplsaa.sb</i>	Contains the OPLS-AA bond stretching and angle bending parameters.
IWAT	watbox	Contains coordinates for equilibrated water boxes for initial system setup.
INAQ1	org1box	Contains coordinates for solvent boxes for non-aqueous solvents.
INAQ2	org2box	Additional solvent boxes for non-aqueous solvents.
ISUM	PLT	The “plt file” receives output coordinates in PDB or mol format for molecular display.
ISUM	SUM	A condensed version of the “ot” file goes to the “sum file” or, for geometry optimizations, the final Z-matrix.

jobs using the provided scripts. Versions of the programs are available to run under UNIX, Linux, and Windows operating systems. The programs are currently distributed by Cemcomco LLC; please see <http://www.Cemcomco.com/> for more information. Job setup by the user focuses on providing the input Z-matrix file; the user also sometimes modifies the “parameter” or “par” file, in which details such as choice of temperature and pressure are specified, and the “command” file, which controls the job execution and items such as the number of FEP windows to run and the lengths of equilibration and averaging for MC calculations. Once the Z-matrix is ready, given the provided scripts and test jobs as models, it is usually possible to set up a new BOSS/MCPRO run in a few minutes or less.

In all, the programs use 15 files, which are described briefly in Table 2. Most of the files are either output by the program or are auxiliary files that contain force-field parameters or preequilibrated solvent boxes. In some cases such as a gas-phase optimization, many of the files are not used. The MC simulations are carried out in a series of runs of ca. 1 million configurations each. The command file specifies the number of runs for the current job. The program is restarted between runs, the current coordinates are in the “in” file, and the results from each run are stored in the “up” file. Current averages for computed quantities are printed in the “ot” file at the end of each run. The existence of the up and in files allows easy extension of any MC simulation.

Command File

The UNIX command file or Windows *bat* file provides the system instructions for the current job. A complete MC simulation for a solute in a solvent can be executed by the UNIX command file in Figure 5. The “setenv” commands make the file assignments and each “time” command causes execution of one MC run of the BOSS program. The only direct input to BOSS from the command

file is three single-integer variables following \$boss, for example, the 111, then the number of configurations for the run (1 million during equilibration), and the lambda values for the reference and two perturbed solutes for an FEP calculation. The latter are all zero for a non-FEP calculation, as in Figure 5. The “foreach blocks” are used here to execute five equilibration runs followed by five averaging runs from this one job submission. The three single-integer variables are ICALC, NEWRUN, and IPRNT. ICALC declares the type of calculation as a continuation of a MC simulation (0), initial setup of a MC simulation (1), gas-phase energy minimization (2), energy scan such as dihedral driving (3), normal mode analysis (4), MC conformational search (5), stochastic conformational search (6), and single-point MM calculation (7). NEWRUN = 1 designates that the averaging is to be restarted beginning with the current MC run, while NEWRUN = 0 causes the averaging to continue. IPRNT controls the verbosity of output to the ot file. The command file for an energy minimization is very simple; after the “setenv” commands, all that is needed is “time \$boss 211 0 0.0 0.0 0.0.”

Parameter File

The par file begins with control variables for the present calculation such as the choice of optimizer for an energy minimization, QM method, the temperature and pressure for a MC simulation, cutoff distances for the potential functions, choices for distribution functions, etc. Short descriptions are given on alternate lines and are mostly self-explanatory. An auxiliary program, *pargen*, is provided with BOSS to create the par file for a new problem; it asks the user a series of questions about the calculation and outputs the appropriate par file. Otherwise, one can copy an old par file and edit it for a new job. In the command files, the *oplsaa.par* file, which contains the OPLS nonbonded and torsional parameters, is


```

#####
echo '          MC for a Solute in a Solvent      '
echo '          25 C & 1 atm '
echo '          Equilibration 5 M/ Averaging 25 M '
#####
#!/bin/csh -f
set boss = $BOSSdir/BOSS
set lambda = "0.000 0.000 0.000"
setenv INFILE      in
setenv UPFILE      up
setenv SAVE        sv
setenv AVERAGE     av
setenv ZMATRIX      zmat
setenv SLVZMAT      slvzmat
setenv SUMMARY      sum
setenv BANGPAR      $BOSSdir/oplsaa.sb
setenv WATERBOX     $BOSSdir/watbox
setenv ORG1BOX      $BOSSdir/org1box
setenv ORG2BOX      $BOSSdir/org2box
cat $BOSSdir/oplsaa.par >> NPTpar
setenv PARAMETER NPTpar
date

##### Equilibration #####
set configurations = "1000000"
setenv OUTPUT      ota
setenv PLTFILE      plta

time $boss 111 $configurations $lambda
foreach block (b c d eq)
    setenv OUTPUT    ot$block
    setenv PLTFILE    plt$block
    time $boss 001 $configurations $lambda
end

##### Averaging #####
set configurations = "5000000"
setenv OUTPUT      ota
setenv PLTFILE      plta

time $boss 011 $configurations $lambda
foreach block (b c d e25)
    setenv OUTPUT    ot$block
    setenv PLTFILE    plt$block
    time $boss 001 $configurations $lambda
end
date
exit

```

Figure 5. Example of a UNIX command file for a MC simulation of a solute in a solvent. The user just needs to provide the solute Z-matrix *zmat*.

appended to the par file, for example, with the cat statement in Figure 5.

Z-Matrix Files

The Z-matrix files contain the internal-coordinate descriptions for the initial setup of the structures of the solutes and of nonstored “custom” solvent molecules. The Z-matrices can be generated in several ways: (1) manually; (2) using the *xPDBZ*, *xMOLZ*, and *xMOL2Z* scripts starting from a PDB, mol, or mol2 file; (3) using the *pepz* utility for building protein or polynucleotide Z-matrices from an input PDB file or from scratch; or (4) using the *BOMB* (biochemical and organic model builder) program, which creates single molecules or combinatorial libraries by themselves or bound to a biomolecular host.⁶⁰ *pepz* and *BOMB* can provide OPLS-AA

atom-typing, while the *x*-scripts generate Z-matrices corresponding to the OPLS/CM1A force field. More than 1000 Z-matrices for organic molecules, ions, complexes, and drugs are distributed with the programs; these can also be modified or used as models for new systems.

A sample Z-matrix file for a potential FEP conversion of all-atom ethanol to ethyl chloride is given in Figure 1. For optimizations, the final Z-matrix is written to the sum file. There are entries at the top for the nine real atoms and two dummy atoms. Each entry gives, in order, the atom number *i*, a label for printing, the OPLS initial atom type, the OPLS final atom type (which may differ from the initial type for FEP calculations), the atom *j* connected to *i*, the *i*–*j* bond length, the atom *k* making the *i*–*j*–*k* bond angle, the *i*–*j*–*k* bond angle, the atom *l* making the *i*–*j*–*k*–*l* dihedral angle, and the *i*–*j*–*k*–*l* dihedral angle. If there is more than one solute, the start of the atom entries for the next solute is indicated by a single line stating “TERZ” by analogy to PDB format.

The other key items in the Z-matrix are the designations for variable bonds, bond angles, and dihedral angles. For fully flexible ethanol, eight bond lengths, seven bond angles, and six dihedrals angles are declared as variable, which agrees with the $3N - 6 = 21$ internal degrees of freedom. Correct evaluation of the energy also requires contributions from “additional” bond angles and dihedral angles. As a simple example, for ethane, there are nine H–C–C–H dihedral terms needed for OPLS-AA energy evaluation. Depending on how the Z-matrix is written, perhaps only one of these is declared as a “variable” dihedral angle. The remaining “additional” dihedral angles can be listed explicitly or they may be determined automatically by the “AUTO” designations. Furthermore, for every ring, there is a ring-closure bond that also needs to be declared as an “additional bond.” If QM is used for the solute energetics, the “additional terms” are irrelevant and do not need to be declared. For MM calculations, the programs look up the force-field parameters for the bond stretching and angle bending terms in the *oplsaa.sb* file and the nonbonded parameters and the Fourier coefficients for the torsional potentials in the par file. In MC simulations, the sampling ranges for the internal degrees of freedom are determined automatically to yield reasonable acceptance rates. This can be overridden by expanded entries in the Z-matrix. As indicated in Figure 1, the Z-matrix is also the place where any harmonic restraints are specified; the format varies depending on the type of restraint (distance, position, dihedral angle, or NOE). For example, the specification of an NOE constraint consists of the atom pair, R_{high} , R_{low} , and the force constant, *k*, for a flat-bottomed harmonic potential.

Upon an attempted MC move, the solute is translated and rotated, the variable entries are changed randomly within their ranges, and the solute molecule is rebuilt. The atoms are constructed in order, which can affect the sampling efficiency. Note the way the methyl group is treated in Figure 1 for concerted rotation of the three methyl hydrogens by the use of improper dihedral angles for atoms 10 and 11. For a flexible chain, it is also best to build out from the center of a molecule rather than proceeding from one end to the other. This causes fewer atoms to be moved on average by the changes in the dihedral angles. A final point to note is that for FEP mutations of one molecule into

another, the entries in the Z-matrix correspond to $\lambda = 0.0$ (the initial structure) and the corresponding values that change in going to the perturbed structure at $\lambda = 1.0$ are listed under "Geometry Variations." For example, for converting ethanol to methyl chloride, the entry "00050001 1.78" indicates that the length for the O—C bond to atom 5 should change from 1.41 to 1.78 Å as it becomes a Cl—C bond. The one Z-matrix contains all the information needed to construct the reference or perturbed solute or any intermediate form.

Stored Solvent Configurations

Another convenient feature of *BOSS* and *MCPRO* is the flexibility in setting up the initial system. The options are controlled by choices in the par file. Stored, equilibrated boxes with up to 750 molecules are provided for 11 common solvents: water (TIP3P, TIP4P, and TIP5P), methanol, acetonitrile, DMSO, argon, dimethyl ether, THF, propane, methylene chloride, chloroform, and carbon tetrachloride. Solutes can be automatically inserted into the center of these boxes and solvent molecules with serious steric conflicts are removed. Furthermore, with *BOSS* one can input any solvent box created by a pure liquid simulation. For such "custom" solvents, the user provides the Z-matrix (*slvzmat* in Fig. 5) of one molecule of the liquid, which is replicated. The initial solvent box is created from a box of liquid argon scaled to the estimated volume of the custom solvent, with a central atom of the custom solvent molecules coincident with the argon atoms. Any of the stored boxes can also be used as a template to create new boxes containing up to several thousand solvent molecules. In addition, equilibrated water spheres are provided with diameters from 36 to 50 Å containing 800 to 2200 water molecules; these are used for easy setup of MC calculations for droplets or the typical water caps used for simulations of proteins. Another 30-Å droplet that is provided contains 1835 small (2-Å diameter) Lennard-Jones spheres; these are typically used to flood a protein–ligand complex to highlight regions where there is room for optimization of the ligand.

Output

The output in the "ot file" from energy minimizations or normal mode calculations lists, as appropriate, the force-field parameters, optimized structure, dipole and quadrupole moments, energy components, vibrational frequencies, and normal modes. Results from a solvent-accessible surface area (SASA) analysis are also always reported including a breakdown into hydrophobic, hydrophilic, carbon- π , and weakly polar (halogens, P, S) components. The Z-matrix for the optimized structure is written to the sum file. If a force-field optimization is performed with *BOSS* for a complex, an atom–atom breakdown of the interaction energy is provided including the separate Coulomb and Lennard-Jones contributions. Calculations with *MCPRO* also provide a breakdown of all solute–solute interactions into the individual residue–residue components. This readily highlights, for example, the protein residues that contribute the most constructively or unfavorably to the protein–ligand interaction energy. The output from a conformational search

gives the structural details, total energies, and number of times found for each unique conformer. Additional files are also generated in this case containing each conformer in mol format, the optimized *BOSS/MCPRO* Z-matrix for each conformer, and a comma-separated-file listing each conformer's energy and dihedral angles. The mol-format file is conveniently displayed by a graphics program such as Accelrys' *WebLabViewer*, which illustrates the conformers overlaid or separated.

The output for Monte Carlo simulations is in two main sections with verbosity controlled by the IPRNT variable, so many of the following items are optional. Before MC calculations begin or resume, the Z-matrix may be listed along with all utilized force-field parameters. The total energy and its components are recomputed from the coordinates in the in file or computed for the first time otherwise. If it is a continuation run (ICALC = 0), then it is required that the "old energy" in the in file agrees with the "new energy" that is computed at the beginning of the new run within the limits of precision of the computer (at least seven figures). Many other parameters for the simulation are listed along with the all energy terms. The λ values are then noted, followed by the coordinates for the reference solute and, if an FEP calculation is being performed, for the two perturbed solutes. The Monte Carlo calculation begins at this point. When it finishes, the acceptance rate information for attempted volume changes in NPT simulations is printed and then some or all of following output is provided depending on IPRNT:

1. plots showing the history for variable dihedral angles in the solutes,
2. the final total energy and its components,
3. the final coordinates for the solutes,
4. the atom–atom radial distribution functions and their integrals (coordination numbers) that have been requested in the parameter file,
5. the solvent and solute total energy and energy pair distribution functions,
6. the distribution functions for the variable dihedral angles,
7. the record of attempted and accepted moves for each solvent and solute molecule and residue,
8. the full report on the computed thermodynamic results including the averages for each run, every energy component for each run, the total averages and the standard deviations (1 σ) calculated from the fluctuations in the separate averages, and the ΔH and ΔS for the two ΔG s that are computed in FEP calculations from an umbrella sampling procedure, and
9. the residue–residue breakdown of the solute–solute interaction energies and results of the SASA analysis.

A short version of the results is also written to the sum file and consists mostly of the thermodynamic data. In all cases, the final coordinates of the system are written to the plt file in PDB or mol format for easy display with molecular graphics programs. The output PDB file is called *plt.pdb* in most cases. It is routine to run a calculation such as an optimization and then execute scripts such as *xRAS* or *xWL* that immediately display the structure in *plt.pdb* using *Rasmol* or *WebLabViewer*. Illustrations such as those in Figures 2–4 have been made from output *plt.pdb* files.

Acknowledgments

Although the *BOSS*, *MCPRO*, and *pepz* programs have been written primarily by the authors, the following individuals also made valuable contributions, which are gratefully acknowledged: Drs. J. Chandrasekhar, George A. Kaminski, Dongchul Lim, Albert C. Pierce, and Jakob P. Ulmschneider. Many other group members have also made important contributions through development of methodology and the OPLS-AA force field, and through testing, running applications, and providing ideas for enhancements.

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